

Experiment 2: Bayes Decision Theory

Aim

To implement Bayes theorem using the SMS Spam Collection Dataset and classify a given SMS message as Spam or Ham.

Program

```
import pandas as pd

from sklearn.feature_extraction.text import CountVectorizer

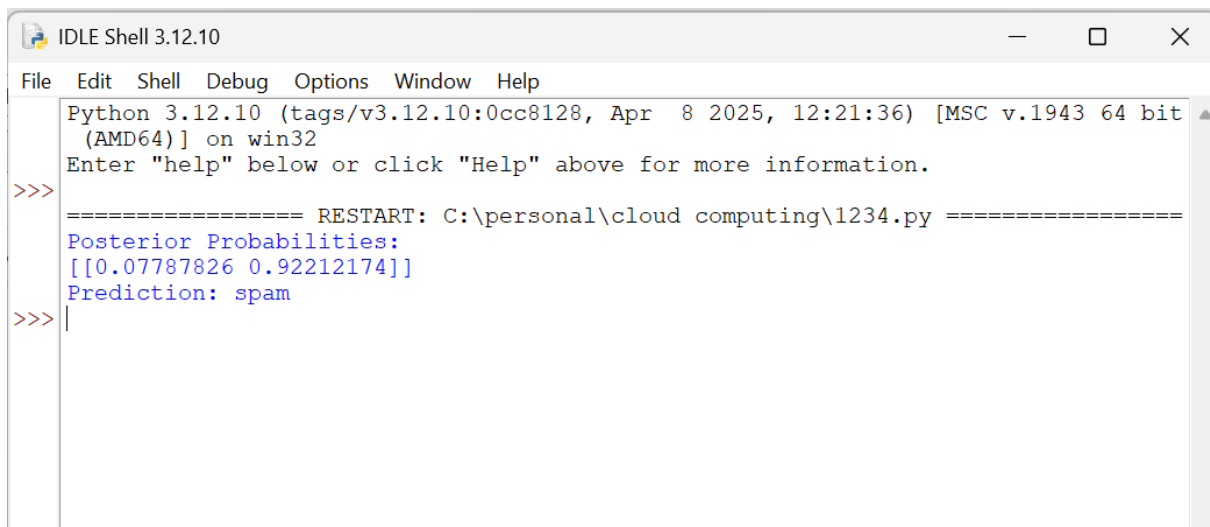
from sklearn.naive_bayes import MultinomialNB

data = {
    "label": [
        "ham", "ham", "ham", "ham", "ham",
        "spam", "spam", "spam", "spam", "spam"
    ],
    "message": [
        "How are you?",
        "Let's meet tomorrow.",
        "Are you coming to class?",
        "Call me when you reach home.",
        "Can we have lunch today?",
        "Congratulations! You won a free lottery.",
        "Claim your cash prize now.",
        "Win a free mobile phone.",
        "Exclusive offer! Click here.",
        "You have won a $1000 gift card."
    ]
}

df = pd.DataFrame(data)
```

```
X = df["message"]
y = df["label"]
vectorizer = CountVectorizer()
X = vectorizer.fit_transform(X)
model = MultinomialNB()
model.fit(X, y)
test_message = ["Congratulations! You won a free ticket."]
test = vectorizer.transform(test_message)
print("Posterior Probabilities:")
print(model.predict_proba(test))
prediction = model.predict(test)
print("Prediction:", prediction[0])
```

Output



```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\personal\cloud computing\1234.py =====
Posterior Probabilities:
[[0.07787826 0.92212174]]
Prediction: spam
>>> |
```

Result

Bayes theorem effectively classified SMS messages using posterior probabilities and achieved accurate spam detection.

